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BSCS C301

Finals Task 03. DAX TIME INTELLIGENCE

Overview:

In this lab, you'll create measures with DAX expressions that involve time intelligence.

In this lab, learn how to:

- Use various time intelligence functions to manipulate filter context that specific concerns dates.

This lab should take approximately 15 minutes.

1. SALES YTD CALCULATION

Formatting

Columns

Calculations

Formatting

Visualizations

Data

1 Sales YTD =

2 TOTALYTD(

3 SUM(Sales[Sales]),

4 DATE[Date],

5 "6-30"

6

7

2020 Oct

2020 Nov

2020 Dec

2021 Jan

2021 Feb

2021 Mar

2021 Apr

2021 May

FY2018

2017 Jul

2017 Aug

2017 Sep

2017 Oct

2017 Nov

2017 Dec

2018 Jan

2018 Feb

2018 Mar

2018 Apr

2018 May

2018 Jun

FY2019

2018 Jul

2018 Aug

2018 Sep

2018 Oct

2018 Nov

2018 Dec

Total

\$748.68

\$419.46

\$475

\$2,146.96

739

8,459

\$16,429,043

\$16,429,042.6

\$655.59

\$419.46

\$519

\$2,146.96

38

352

\$489,328

\$489,328.4

\$758.93

\$419.46

\$475

\$2,146.96

75

785

\$1,540,072

\$2,029,400.42

\$741.85

\$419.46

\$519

\$2,146.96

60

593

\$1,186,332

\$3,195,732.87

\$677.45

\$419.46

\$519

\$2,146.96

40

489

\$844,833

\$4,040,565.58

\$752.31

\$419.46

\$501

\$2,146.96

90

1,106

\$2,325,755

\$6,366,320.49

\$734.58

\$419.46

\$501

\$2,146.96

63

803

\$1,703,435

\$8,069,755.58

\$608.04

\$419.46

\$519

\$2,146.96

40

377

\$713,230

\$8,762,895.12

\$696.80

\$419.46

\$501

\$2,146.96

79

866

\$1,900,794

\$10,683,779.58

\$863.54

\$419.46

\$519

\$2,146.96

64

653

\$1,455,280

\$12,139,059.85

\$732.25

\$419.46

\$519

\$2,146.96

37

494

\$883,011

\$13,022,070.83

\$763.30

\$419.46

\$475

\$2,146.96

85

1,132

\$2,269,720

\$13,291,790.7

\$552.95

\$419.46

\$475

\$2,146.96

68

819

\$1,137,252

\$16,429,042.6

\$397.81

\$202.33

\$432

\$1,466.01

1,255

21,670

\$2,979,780

\$27,979,779.53

\$350.74

\$196.33

\$475

\$1,466.01

72

1,723

\$2,411,559

\$22,411,558.85

\$372.75

\$202.33

\$475

\$1,466.01

139

2,964

\$3,615,914

\$6,027,473.31

\$365.30

\$196.33

\$432

\$1,466.01

111

2,185

\$2,894,647

\$8,922,119.92

\$382.00

\$202.33

\$519

\$1,466.01

133

1,406

\$1,894,177

\$10,726,296.85

\$412.48

\$202.26

\$501

\$1,466.01

133

2,345

\$3,034,497

\$13,781,293.37

\$407.08

\$202.33

\$475

\$1,466.01

114

1,732

\$2,188,206

\$15,969,499.23

\$446.39

\$214.44

\$1,133

\$2,146.96

3,616

57,851

\$7,548,570

0.00%

Build visual

Targets

Date

Product

Region

Reseller

Sales

Cost

Counts

Pricing

Profit

Profit Margin

Quantity

Ratios

Sales YoY Growth

Sales YTD

Salesperson

Salesperson (Performance)

Rows

Year

Month

Columns

Add data fields here

Values

Avg Price

Median Price

Min Price

Max Price

Orders

Order Lines

Sum of Sales

2. SALES YoY Growth (Percentage Logic)

The screenshot shows the Microsoft Power BI Desktop interface. At the top, the title bar indicates the file is named "06-Starter-Sales Analysis [Read-Only] - Power BI Desktop" and was last saved at 10:37 AM. The ribbon menu is set to "Measure tools".

The DAX editor on the left contains the following code:

```

1 Sales YoY Growth =
2 VAR SalesPriorYear =
3     CALCULATE(
4         SUM(Sales[Sales]),
5         PARALLELPERIOD(
6             'Date'[Date],
7             -12,
8             MONTH
9         )
10     )
11 RETURN
12     DIVIDE(
13         (SUM(Sales[Sales]) - SalesPriorYear),
14         SalesPriorYear
15     )
16

```

The data table on the right displays the following values:

	2017 Jul	2017 Aug	2017 Sep	2017 Oct
Sales	\$655.59	\$758.93	\$741.85	\$677.45
Sales Prior Year	\$419.46	\$419.46	\$419.46	\$419.46
Sales YoY Growth	55.19%	80.75%	75.93%	61.83%

3. Final Output

Year	Avg Price	Median Price	Min Price	Max Price	Orders	Order Lines	Sum of Sales	Sales YTD	Sales YoY Growth
FY2021									-100.00%
2020 Jul									-100.00%
2020 Aug									-100.00%
2020 Sep									-100.00%
2020 Oct									-100.00%
2020 Nov									-100.00%
2020 Dec									-100.00%
2021 Jan									-100.00%
2021 Feb									-100.00%
2021 Mar									-100.00%
2021 Apr									-100.00%
2021 May									-100.00%
FY2018	\$748.68	\$419.46	\$4.75	\$2,146.96	739	8,459	\$16,429,043	₱16,429,042.6	
2017 Jul	\$655.59	\$419.46	\$5.19	\$2,146.96	38	352	\$489,328	₱489,328.4	
2017 Aug	\$758.93	\$419.46	\$4.75	\$2,146.96	75	785	\$1,540,072	₱2,029,400.42	
2017 Sep	\$741.85	\$419.46	\$5.19	\$2,146.96	60	593	\$1,166,332	₱3,195,732.87	
2017 Oct	\$677.45	\$419.46	\$5.19	\$2,146.96	40	499	\$844,833	₱4,040,565.58	
2017 Nov	\$752.31	\$419.46	\$5.01	\$2,146.96	90	1,106	\$2,325,755	₱6,366,320.49	
2017 Dec	\$734.58	\$419.46	\$5.01	\$2,146.96	63	803	\$1,703,435	₱8,069,755.58	
2018 Jan	\$808.94	\$419.46	\$5.19	\$2,146.96	40	377	\$713,230	₱8,782,985.12	
2018 Feb	\$896.80	\$419.46	\$5.01	\$2,146.96	79	866	\$1,900,794	₱10,683,779.58	
2018 Mar	\$863.54	\$419.46	\$5.19	\$2,146.96	64	653	\$1,455,280	₱12,139,059.85	
2018 Apr	\$732.25	\$419.46	\$5.19	\$2,146.96	37	494	\$883,011	₱13,022,070.83	
2018 May	\$761.30	\$419.46	\$4.75	\$2,146.96	85	1,112	\$2,269,720	₱15,291,790.7	
2018 Jun	\$552.95	\$419.46	\$4.75	\$2,146.96	68	819	\$1,137,252	₱16,429,042.6	
FY2019	\$397.81	\$202.33	\$4.32	\$1,466.01	1,255	21,670	\$27,979,780	₱27,979,779.53	70.31%
2018 Jul	\$350.74	\$196.33	\$4.75	\$1,466.01	72	1,723	\$2,411,559	₱2,411,558.85	392.83%
2018 Aug	\$372.75	\$202.33	\$4.75	\$1,466.01	139	2,964	\$3,615,914	₱6,027,473.31	134.79%
2018 Sep	\$365.30	\$196.33	\$4.32	\$1,466.01	111	2,185	\$2,894,647	₱8,922,119.92	148.18%
2018 Oct	\$382.00	\$202.33	\$5.19	\$1,466.01	73	1,406	\$1,804,177	₱10,726,296.85	113.55%
2018 Nov	\$412.48	\$209.26	\$5.01	\$1,466.01	133	2,345	\$3,054,997	₱13,781,293.57	31.36%
2018 Dec	\$407.08	\$202.33	\$4.75	\$1,466.01	114	1,732	\$2,188,206	₱15,969,499.23	28.46%
Total	\$446.39	\$214.24	\$1.33	\$2,146.96	3,616	57,851	\$77,548,570		0.00%